

Diffusion Policy for Multi-Step Block Assembly

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GitHub Colab Notebook

1 Introduction

Diffusion Policy [1] has demonstrated strong performance on visuomotor control tasks by formulating action prediction as a conditional denoising diffusion process. We implement and evaluate this approach on a multi-step block assembly task using a single Franka Emika Panda robot.

Multi-step tasks require sequential subtasks with temporal dependencies. Unlike atomic tasks (single pick-and-place) addressed by models like NVIDIA’s GR00T N1, multi-step tasks involve sequential composition where early actions affect later outcomes. Our task—2–3 consecutive pick-and-place operations—extends the original Diffusion Policy work to evaluate performance on longer action sequences.

2 Task Description

Setup: Universal Robot (UR) 10e, 3D-printed template with 2–3 slots, cubic blocks (50-70 mm in width and length), ArUco markers on cubes and slots, overhead camera.

Task: Sequentially grasp each cube and insert into corresponding slots. This tests contact-rich manipulation, precise 6-DOF alignment, and sequential decision-making over 8–12 action chunks.

3 Modification Description

3.1 Experimental Conditions

Primary: State-Based Policy Low-dimensional observations from ArUco markers: cube poses $(x, y, z, q) \in R^{7 \times N}$ where $N = 2$ or 3 , slot poses, end-effector pose, gripper state.

Advantages: faster training convergence, 30–50 demos sufficient, easier debugging, direct 6-DOF information eliminates perception bottleneck.

Extension (if time): Vision-Based Policy RGB images $(640 \times 480 \times 3)$ with ResNet-18 encoder pre-trained on ImageNet, proprioception (joint angles), gripper state.

Advantages: marker-free operation, better generalization to visual variations. Requires 80–120 demos due to higher-dimensional input.

3.2 Policy Architectures and Implementation

Monolithic Policy: Single diffusion policy observing all objects, outputting action sequences for complete assembly. Uses temporal CNN for observation history ($T_{obs} = 2$) and predicts action chunks ($T_{action} = 16$). Enables end-to-end task sequencing learning.

We follow DDPM with $K = 100$ diffusion steps. Action horizon prediction with overlapping chunks provides temporal smoothness and implicit closed-loop control through replanning at each timestep.

4 Motivation

Sequential Composition: Most Diffusion Policy work focuses on single-skill tasks. Multi-step tasks face error accumulation and maintaining context over extended sequences. Our 8–12 action chunk task tests coherent behavior without explicit hierarchical decomposition.

Sample Efficiency: State-based observations provide direct pose information, allowing us to study how sample efficiency scales with task complexity and number of demonstrations.

Implicit Skill Learning: Foundation models like GR00T N1 excel at atomic skills but composing them into multi-step behaviors remains challenging. We investigate whether end-to-end imitation learning can implicitly learn composition.

Technical Contributions: Multi-step manipulation benchmark with open-source code; quantitative analysis of demonstration requirements; failure mode characterization; hyperparameter ablations for sequential tasks.

Feasibility

The proposed modification is achievable within 6 weeks using available resources:

Timeline: W1–2: Setup and teleoperation; W3: collect 30–50 demos, train policy; W4: evaluate and tune; W5: additional demos/ablations; W6: final tests and report.

Resources & Risk Mitigation: 3D-printed cubes with chamfers and multi-face markers, Franka Panda robot, and state-based policy reduce complexity and improve sample efficiency. Force-torque monitoring ensures safety. Start with 2 blocks, expand to 3 if stable.



Figure 1: UR10e



Figure 2: 3D Printed Cubes with markers.

References

- [1] Chi et al. *Diffusion Policy: Visuomotor Policy Learning via Action Diffusion*. RSS 2023.
- [2] Dalal et al. *Plan-Seq-Learn: Language Model Guided RL for Long Horizon Tasks*. arXiv:2405.01534, 2024.
- [3] NVIDIA. *Isaac GR00T N1: Foundation Model for Humanoid Robots*. 2025.